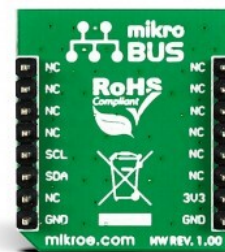
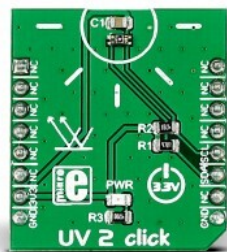


UV 2 Click



PID: MIKROE-2378

UV 2 Click is a compact add-on board that measures ultraviolet light. This board features the [VEML6075](#), a UVA and UVB light sensor with an I2C interface from [Vishay](#). It incorporates a photodiode, amplifiers, and analog/digital circuits into a single device using CMOS technology and converts solar UV light intensity to digital data in 16-bit resolution. The sensor has a specified peak UVA sensitivity of 365nm and a peak UVB sensitivity of 330nm. It also has temperature compensation capabilities to keep a stable output regardless of the temperature conditions. This Click board™ makes the perfect solution for the development of applications based on systems that warn of excessive UV exposure or industrial applications where UV is used for sterilization.

UV 2 Click is supported by a [mikroSDK](#) compliant library, which includes functions that simplify software development. This [Click board™](#) comes as a fully tested product, ready to be used on a system equipped with the [mikroBUS™](#) socket.

How does it work?

UV 2 Click is based on the VEML6075, a UVA and UVB light sensor with an I2C interface from Vishay. The UVA and UVB have their own individual channels, along with the UVD as a dummy channel for dark current cancellation, and UVcomp1, and UVcomp2, parts of a normalized spectral response. All those values are essential for deriving the UV radiation values from the sensor readings. The measurement results are stored in separate registers. They remain in registers and can be read from them until the device wakes up and a new measurement is made.

The UVB rays wavelengths ranging from 280nm to 320nm are extremely energetic and harmful

Mikroe produces entire development toolchains for all major microcontroller architectures.

Committed to excellency, we are dedicated to helping engineers bring the project development up to speed and achieve outstanding results.



ISO 27001: 2013 certification of informational security management system.
ISO 14001: 2015 certification of environmental management system.
OHSAS 18001: 2008 certification of occupational health and safety management system.



ISO 9001: 2015 certification of quality management system (QMS).

for the skin to the extent that they are responsible for 65% of skin tumors. Thankfully, only 0.1% of the solar energy that arrives on the earth's surface is in the shape of UVB radiation.

The UVA ray wavelengths ranging from 320nm to 400nm are less powerful than the previous ones but highly penetrating. They can reach the skin, becoming responsible for photoaging and promoting the onset of different forms of skin cancer. 4.9% of solar energy is made up of UVA rays. The UV 2 Click communicates with the host MCU using an I2C interface over the mikroBUS™ socket, supporting Standard (100KHz) and Fast (400KHz) operating frequencies.

This Click board™ can only be operated with a 3.3V logic voltage level. The board must perform appropriate logic voltage level conversion before using MCUs with different logic levels. However, the Click board™ comes equipped with a library containing functions and an example code that can be used as a reference for further development.

Specifications

Type	Optical
Applications	Can be used for the development of applications based on systems that warn of excessive UV exposure, or industrial applications where UV is used for sterilization
On-board modules	VEML6075 - UVA and UVB light sensor with an I2C interface from Vishay
Key Features	Excellent UVA, and UVB peak sensitivities, reliable performance of UV radiation measurement under long-time solar exposure, integrated UV sensor, and signal conditioning IC, low power consumption, 16-bit resolution per channel, converts solar UV signal to digital data, and more
Interface	I2C
Feature	No ClickID
Compatibility	mikroBUS™
Click board size	S (28.6 x 25.4 mm)
Input Voltage	3.3V
Category	Click Boards

Pinout diagram

This table shows how the pinout on 9DOF 2 Click corresponds to the pinout on the mikroBUS™ socket (the latter shown in the two middle columns).

Notes	Pin					Pin	Notes
	NC	1	AN	PWM	16	NC	
	NC	2	RST	INT	15	NC	
	NC	3	CS	RX	14	NC	

Mikroe produces entire development toolchains for all major microcontroller architectures.

Committed to excellency, we are dedicated to helping engineers bring the project development up to speed and achieve outstanding results.



ISO 27001: 2013 certification of informational security management system.
ISO 14001: 2015 certification of environmental management system.
OHSAS 18001: 2008 certification of occupational health and safety management system.



ISO 9001: 2015 certification of quality management system (QMS).

	NC	4	SCK	TX	13	NC	
	NC	5	MISO	SCL	12	SCL	I2C Clock
	NC	6	MOSI	SDA	11	SDA	I2C Data
Power Supply	3.3V	7	3.3V	5V	10	NC	
Ground	GND	8	GND	GND	9	GND	Ground

Onboard settings and indicators

Label	Name	Default	Description
LD1	PWR	-	Power LED Indicator

UV2 Click electrical specifications

Description	Min	Typ	Max	Unit
Supply Voltage	-	3.3	-	V
Peak UVA Sensitivity	-	-	365	nm
Peak UVB Sensitivity	-	-	330	nm
Range of Spectral Bandwidth	-	±10	-	nm

Software Support

We provide a library for the UV 2 Click as well as a demo application (example), developed using MIKROE [compilers](#). The demo can run on all the main MIKROE [development boards](#).

Package can be downloaded/installed directly from NECTO Studio Package Manager(recommended), downloaded from our [LibStock™](#) or found on [Mikroe github account](#).

Library Description

This library contains API for UV 2 Click driver.

Key functions

- This function set active force mode by write force mode UV_AF bit to config register of VEML6075 sesnor on UV 2 Click.
- This function get UVA data by read UVA register value of VEML6075 sesnor on UV 2 Click.
- This function get UVB data by read UVB register value of VEML6075 sesnor on UV 2 Click.

Example Description

This app measurement UVA and UVB data and calculate UV index level.

The full application code, and ready to use projects can be installed directly from NECTO Studio Package Manager(recommended), downloaded from our [LibStock™](#) or found on [Mikroe github account](#).

Other Mikroe Libraries used in the example:

Mikroe produces entire development toolchains for all major microcontroller architectures.

Committed to excellency, we are dedicated to helping engineers bring the project development up to speed and achieve outstanding results.



ISO 27001: 2013 certification of informational security management system.
ISO 14001: 2015 certification of environmental management system.
OHSAS 18001: 2008 certification of occupational health and safety management system.



ISO 9001: 2015 certification of quality management system (QMS).

- MikroSDK.Board
- MikroSDK.Log
- Click.UV2

Additional notes and informations

Depending on the development board you are using, you may need [USB UART click](#), [USB UART 2 Click](#) or [RS232 Click](#) to connect to your PC, for development systems with no UART to USB interface available on the board. UART terminal is available in all MIKROE [compilers](#).

mikroSDK

This Click board™ is supported with [mikroSDK](#) - MIKROE Software Development Kit. To ensure proper operation of mikroSDK compliant Click board™ demo applications, mikroSDK should be downloaded from the [LibStock](#) and installed for the compiler you are using.

For more information about mikroSDK, visit the [official page](#).

Resources

[mikroBUS™](#)

[mikroSDK](#)

[Click board™ Catalog](#)

[Click Boards™](#)

Downloads

[UV 2 click example on Libstock](#)

[UV 2 click Documentation](#)

[VEML6075 datasheet](#)

[UV2 click 2D and 3D files](#)

[UV2 click schematic](#)

Mikroe produces entire development toolchains for all major microcontroller architectures.

Committed to excellency, we are dedicated to helping engineers bring the project development up to speed and achieve outstanding results.



ISO 27001: 2013 certification of informational security management system.
 ISO 14001: 2015 certification of environmental management system.
 OHSAS 18001: 2008 certification of occupational health and safety management system.



ISO 9001: 2015 certification of quality management system (QMS).